## Question 18 — 1 mark

If $P(3,5)$ is a point on the graph of $y = f(x)$, identify the corresponding point on the graph of $y = f(x-1) + 7$.

a) $(2,12)$

b) $(4,-2)$

c) $(2,-2)$

d) $(4,12)$

## Question 19 — 1 mark

Identify how the graph of $y = 3^x$ is transformed to the graph of $y = 3^{-x}$.

a) reflected over the $x$-axis

b) reflected over the $y$-axis

c) reflected over both the $x$-axis and the $y$-axis

d) reflected over the line $y = x$

## Question 20 — 1 mark

Identify the equation $\log_a b = c$ in exponential form.

a) $b^c = a$

b) $a^c = b$

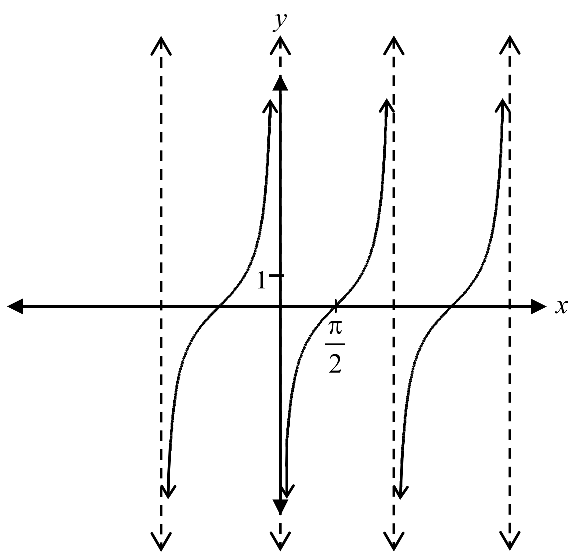
c) $a^b = c$

d) $c^a = b$

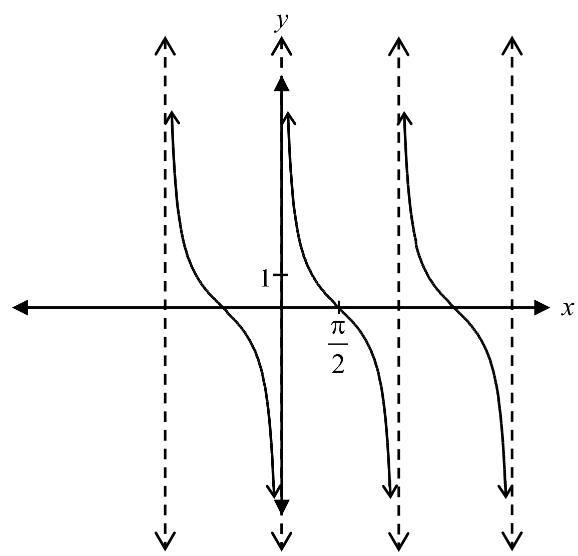
## Question 21 1 mark

Identify the graph of $y = \tan x$.

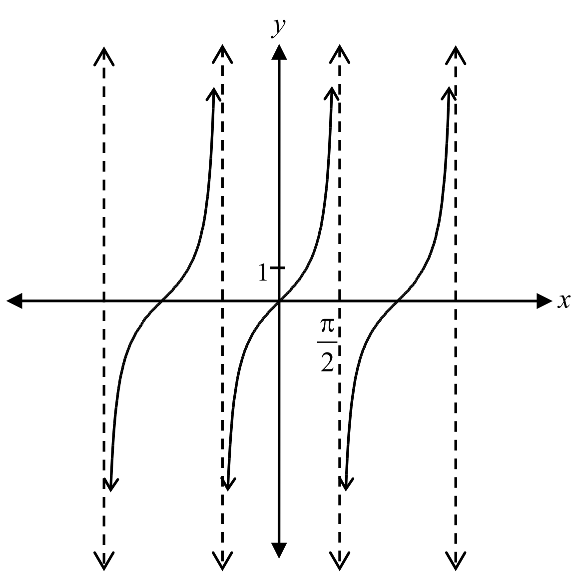
a)



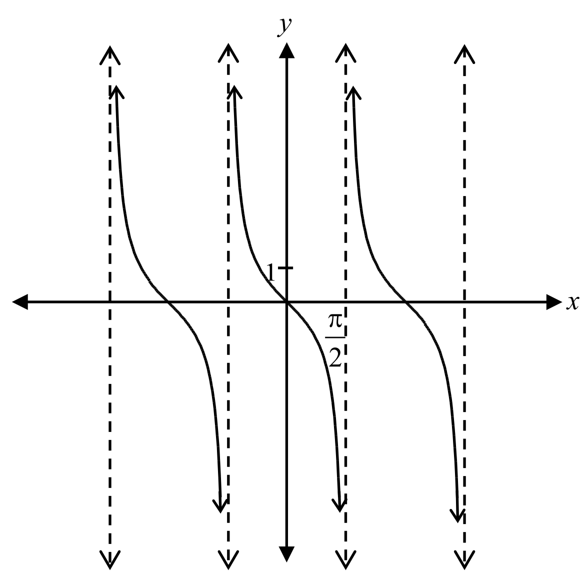

b)




c)



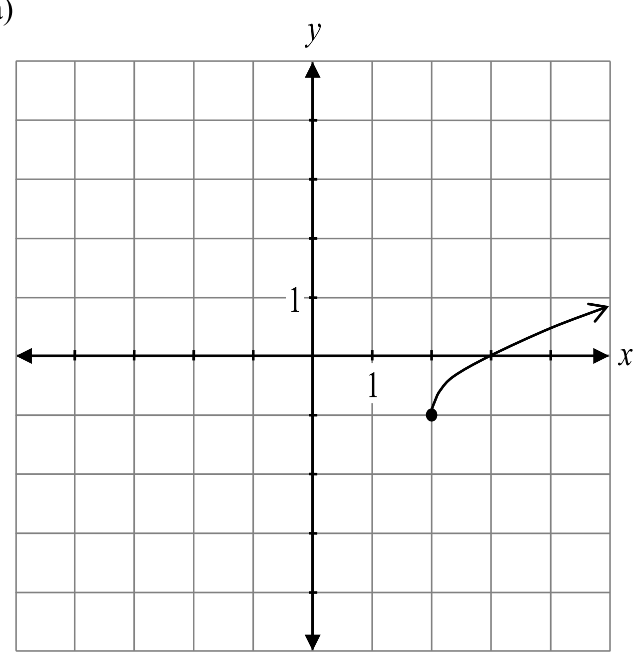

d)



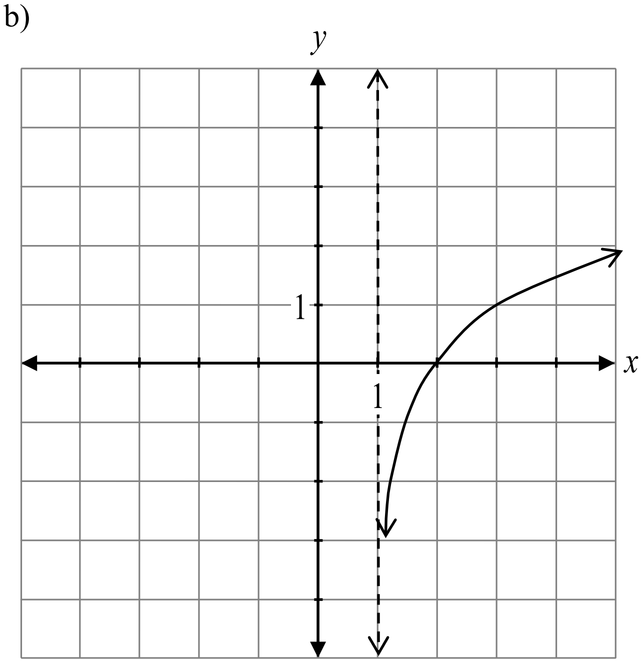
## Question 22 1 mark

Identify which of the following graphs represents a logarithmic function.

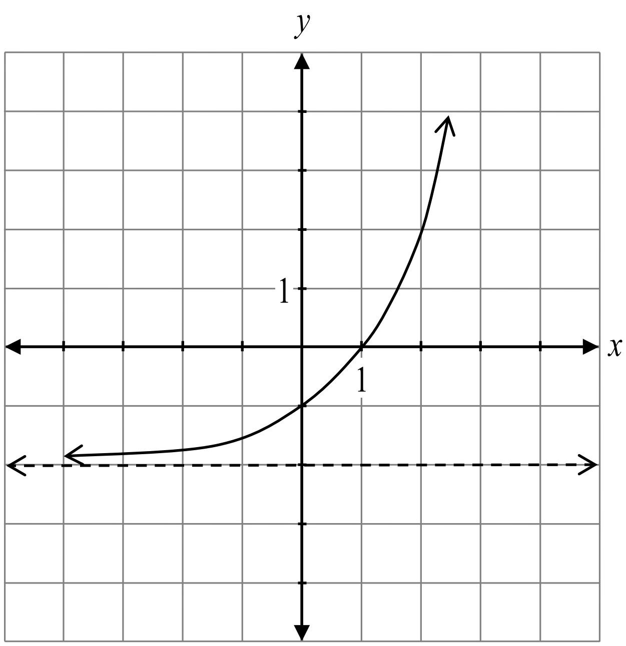
a)



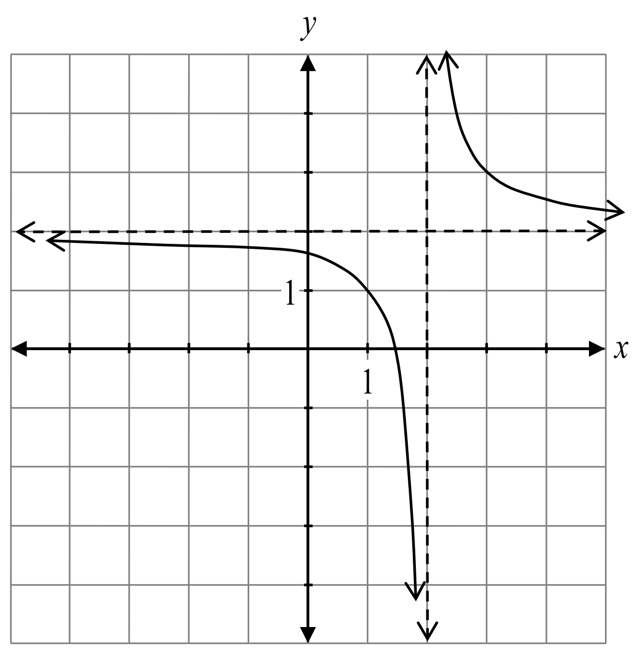

b)




c)




d)



## Question 23 — 1 mark

If the volume of a box is represented by $V(x) = (x+4)(x+2)(x-1)$, identify a possible value of $x$.

a) $-4$

b) $-1$

c) $1$

d) $4$

## Question 24 — 1 mark

Identify a coterminal angle for $\theta = -\frac{\pi}{3}$.

a) $\frac{\pi}{3}$

b) $\frac{4\pi}{3}$

c) $\frac{7\pi}{3}$

d) $\frac{11\pi}{3}$

## Question 25 — 1 mark

Identify the value of $n$ in the equation ${}_{n}C_{3} = {}_{n}C_{6}$.

a) 3

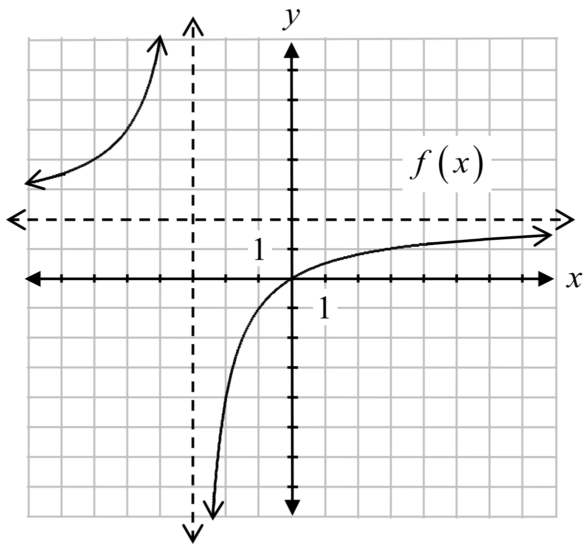
b) 6

c) 9

d) 18

## Question 26 1 mark

Identify the equation of the function, $f(x)$, for the following graph.




a) $f(x) = \dfrac{2x}{x+3}$

b) $f(x) = \dfrac{2}{x+3}$

c) $f(x) = \dfrac{2x^2}{x(x+3)}$

d) $f(x) = \dfrac{3x^2}{x(x+2)}$

## Question 27 — 2 marks

Match the following radical functions with their graphs.

Place the appropriate letter in this column.

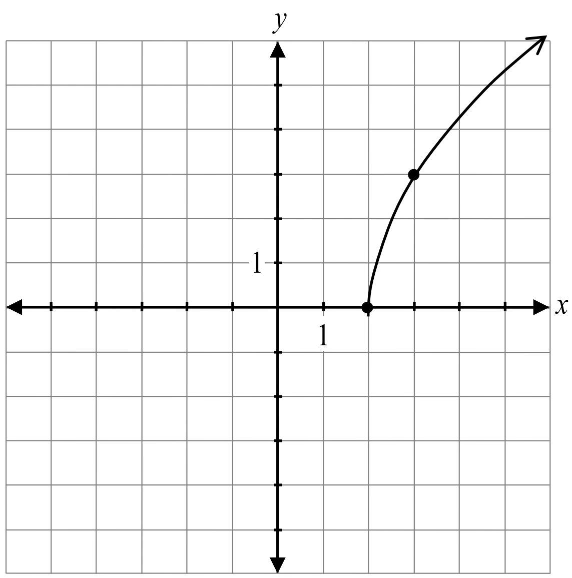
$f(x) = 2\sqrt{-(x+3)}$ ________

$g(x) = -2\sqrt{(x+3)}$ ________

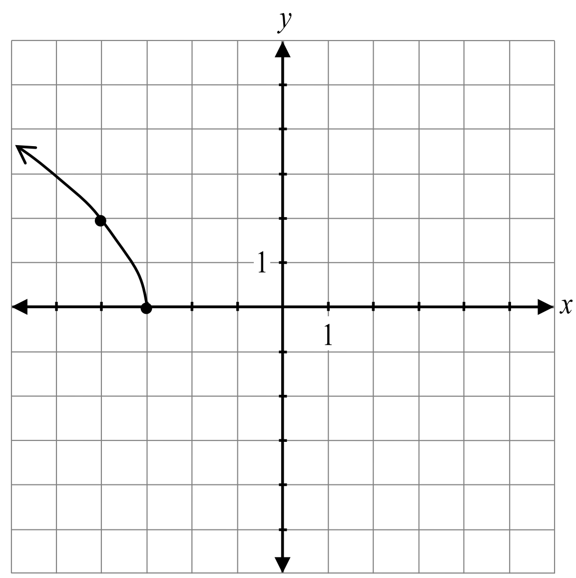
$h(x) = 3\sqrt{(x-2)}$ ________

$k(x) = \sqrt{3(x-2)}$ ________

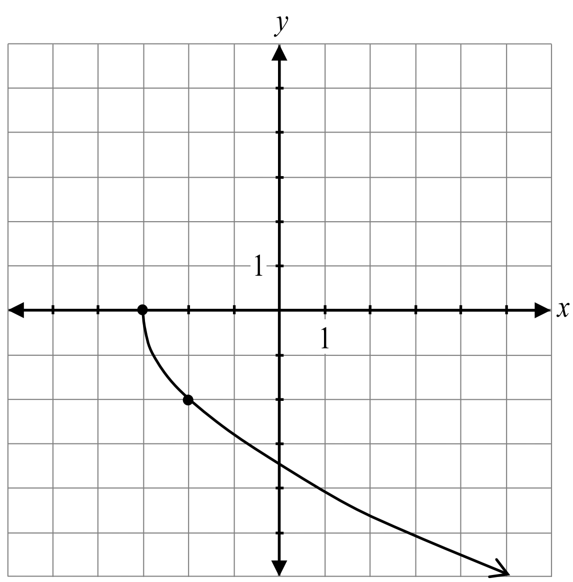
A)



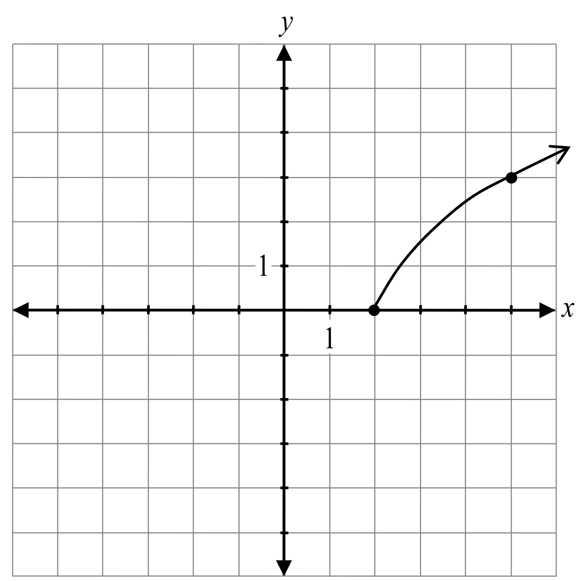

B)




C)




D)



## Question 28 3 marks 119

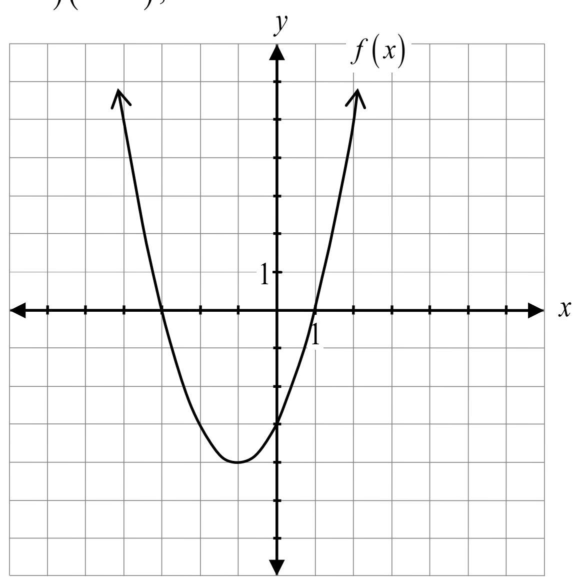
Express $p(x) = x^3 - 2x^2 - 4x + 8$ as a product of factors.

$p(x) =$ ______________________________

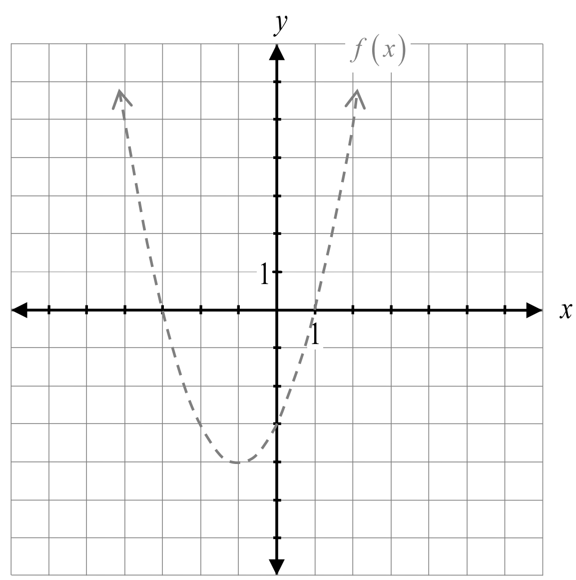
## Question 29

a) 3 marks b) 1 mark 120 121

Given the graph of $f(x)=(x+3)(x-1)$,




a) sketch the graph of $g(x)=\dfrac{1}{f(x)}$.




The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

b) describe how to sketch the graph of $h(x)=|f(x)|$.

## Question 30 — 1 mark 122

Describe how the value of $m$ in the equation $y = \log_3(x - m)$, $m \in \mathbb{R}$, affects the asymptote on the graph of $y = \log_3 x$.

## Question 31 — 2 marks 123

Solve algebraically.

$$25^x = \left(\frac{1}{5}\right)^{-3x+1}$$

## Question 32 4 marks 124

Solve $\cos 2\theta = 0$, where $\theta \in \mathbb{R}$.

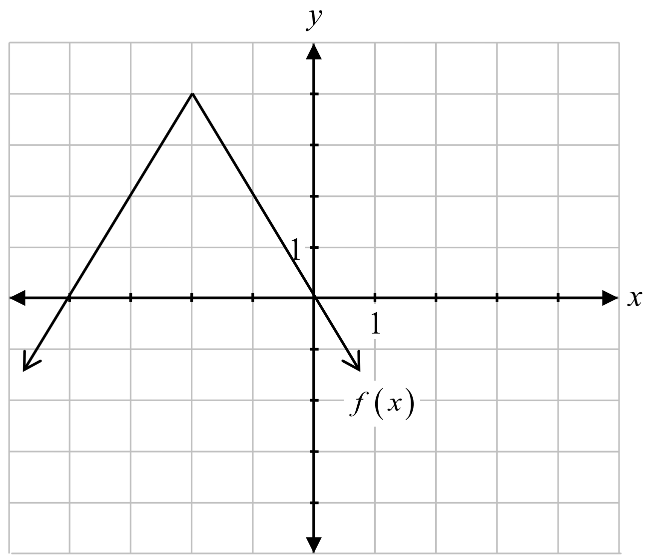
## Question 33 1 mark

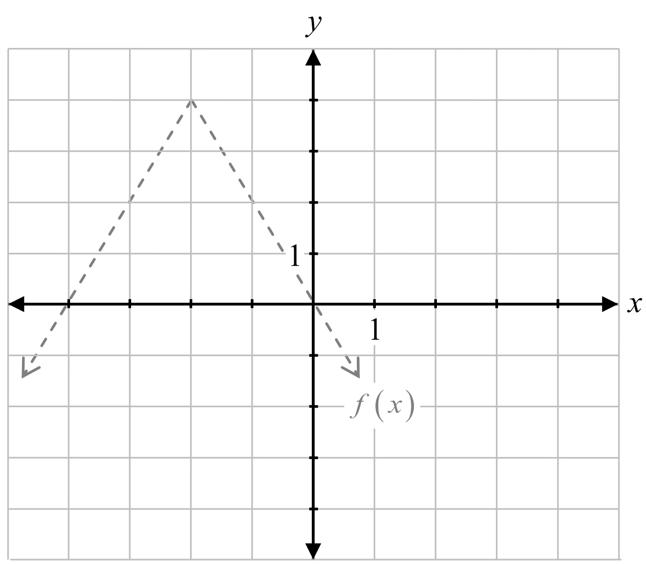


Describe a difference between the graphs of $y = f(x)$ and $y = g(x)$.

$$f(x) = -2(x+1)^2(x+3)$$
$$g(x) = 2(x+1)^2(x+3)$$

Given the graph of $y = f(x)$, sketch the graph of $\sqrt{f(x)}$.




The graph of $f(x)$ has already been drawn for your reference.

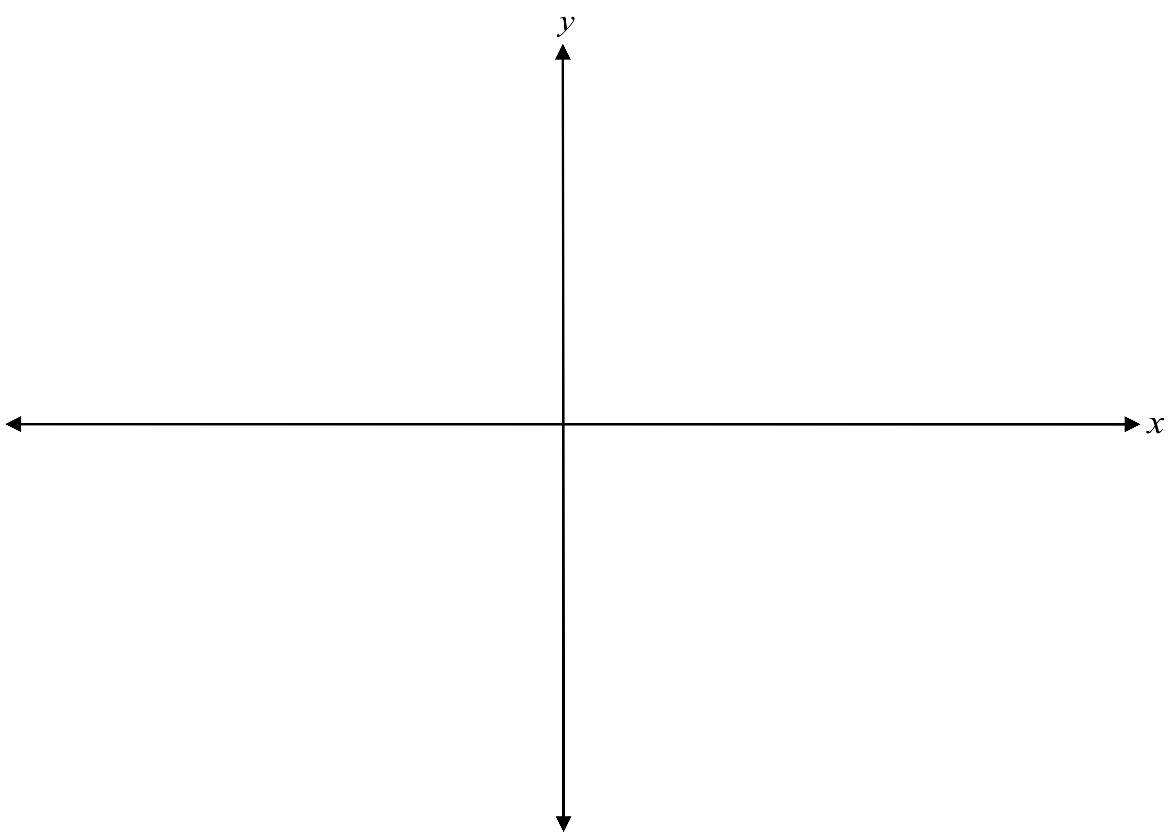
No marks will be awarded for the graph of $f(x)$.

## Question 35 1 mark 127

Describe the relationship between the zeros of the function $f(x) = (2x-1)(x+3)^2$, the roots of the equation $(2x-1)(x+3)^2 = 0$, and the $x$-intercepts of the graph of $y = f(x)$.

## Question 36 3 marks 128

Sketch a graph of at least one period of the function $f(x) = \cos\left[\frac{1}{2}\left(x + \frac{\pi}{2}\right)\right] - 3$.



## Question 37 — 1 mark

129

Verify that $\theta = \frac{4\pi}{3}$ is a solution of the equation $4\cos^2\theta - 1 = 0$.

## Question 38 — 1 mark 130

Describe how to determine the equation of the horizontal asymptote of a rational function when the degree of the polynomial in the numerator and the degree of the polynomial in the denominator are equal.

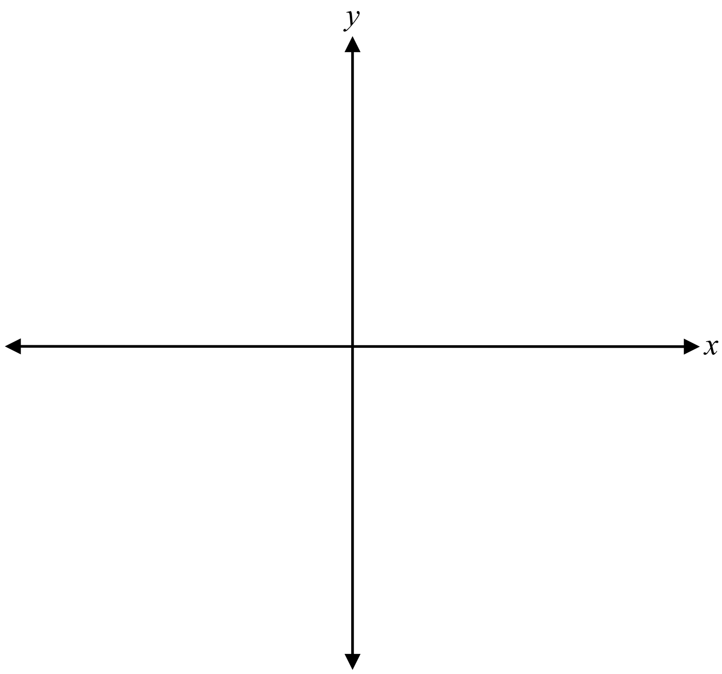
## Question 39 — 2 marks 131

Evaluate.

$$\frac{\cot\left(-\frac{5\pi}{6}\right)}{\sin\left(\frac{17\pi}{3}\right)}$$

## Question 40 — 4 marks 132

Sketch the graph of the function $f\left(x\right)=\dfrac{-1}{\left(x-1\right)^{2}}$ and determine the range.




Range: ________________________________

## Question 41 a) 1 mark b) 1 mark 133 134

Given $f(x) = \sqrt{x-2}$ and $g(x) = x^2 + 1$,

a) determine $g(f(x))$.

$g(f(x)) =$ ______________________________

b) explain why the domain of $g(f(x))$ is restricted.

## Question 42 3 marks 135

Solve algebraically.

$$2\log_a 3 + \log_a 4 = 2, \text{ where } a > 0$$

## Question 43 2 marks 136

Solve $\sec\theta + 2 = 0$ over the interval $[0,\ 2\pi]$.

## Question 44 1 mark 137

Determine the *x*-intercept of the graph of $f(x) = e^x - 1$.

## Question 45 1 mark 138

Given the $5^{th}$ row of Pascal's triangle, determine the values of the next row.

1 4 6 4 1

## Question 46 2 marks 139

Evaluate.

$$\log_2 80 - \log_2 10$$

## Question 47 1 mark 140

State the amplitude of $f(x) = -2\sin(x - \pi) - 1$.

## Question 48 3 marks 141

Determine the exact value of $\cos 15^{\circ}$.

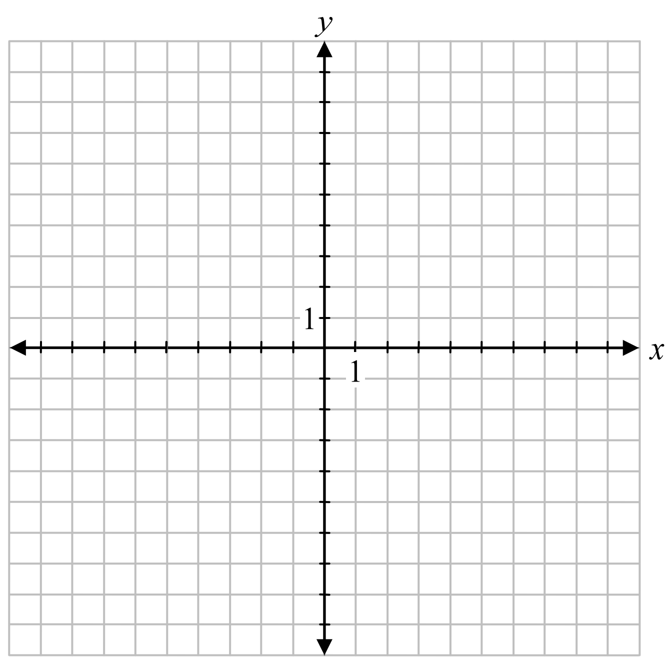
## Question 49 a) 1 mark b) 1 mark 142 143

Given $f(x) = x^2 + 5x + 6$, $g(x) = x + 3$, and $h(x) = f(x) - g(x)$,

a) determine $h(x)$.

$h(x) =$ ____________________

b) sketch the graph of $y = h(x)$.



No marks will be awarded for work done on this page.